

Two years after ChatGPT: a thematic analysis of First-Year students' reflections on AI tool use in higher education

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Abstract

Purpose – This study aims to investigate how first-year university students engage with generative AI tools, especially ChatGPT, in real academic contexts. As AI becomes more embedded in higher education, it is critical to understand how students use these technologies, what benefits they perceive and what concerns they raise. By analyzing 38 student reflections from a Fundamentals of Technology course, the study explores student motivations, behaviors and attitudes, contributing to a more grounded understanding of how generative AI is reshaping undergraduate learning practices.

Design/methodology/approach – Using an inductive thematic analysis, this qualitative study examines 38 voluntary student reflections on AI tool use in a first-year technology course. Thematic analysis followed Braun and Clarke's six-phase process and was grounded in students' authentic descriptions of their interactions with AI. No predefined codebook was used. Themes emerged around the purposes for which students used AI, perceived benefits, challenges and attitudes toward the technology. This bottom-up approach ensured findings remained closely tied to student voices and everyday academic experience.

Findings – Five major themes emerged: (1) purpose of AI use, (2) perceived benefits, (3) challenges with AI, (4) student attitudes and (5) specific AI tools used. Students reported using AI for writing support, problem-solving and idea generation. ChatGPT was the most frequently used tool, followed by Grammarly, QuillBot and Perplexity. While most reflections were positive, some students expressed concerns about overreliance, privacy and accuracy. The findings suggest students view AI as a flexible academic aid, but with limited reflection on potential drawbacks or ethical considerations.

Research limitations/implications – This study is limited by its sample size and context, focusing on a single course at one institution. Reflections were not initially collected for research purposes, which may impact depth and generalizability. However, the candid, realworld insights provide valuable starting points for understanding AI integration in early undergraduate education. Future research should explore AI use across diverse institutions and instructional contexts and investigate how students develop AI literacy and ethical reasoning over time.

Practical implications – As generative AI becomes commonplace in student workflows, educators must rethink assignment design, academic integrity policies and digital literacy instruction. This study suggests that many students already use AI for legitimate academic support, such as writing organization and math problem-solving. Rather than prohibiting these tools, faculty can integrate them into coursework, encouraging critical reflection on their use. Introducing prompt engineering and source evaluation into curricula may help students use AI responsibly while enhancing their learning outcomes.

Social implications – Student reflections indicate that AI tools like ChatGPT may help democratize academic support by offering accessible, on-demand assistance, particularly for those without access to tutoring or support services. However, concerns around data privacy, overreliance and potential impacts on



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independent thinking underscore the need for institutional guidance. These findings highlight broader tensions between innovation, equity and ethics in higher education's digital transformation.

Originality/value – This paper offers a rare, grounded analysis of real student reflections on AI tool use in academic settings. While much existing research is hypothetical or policy-driven, this study centers student voices, providing insights into actual behaviors, motivations and concerns. By focusing on first-year students, the study reveals early patterns in AI adoption and contributes to conversations about ethical, instructional and infrastructural responses in higher education.

Keywords Education, Technology in the classroom, Academic integrity, AI adoption, ChatGPT, Generative AI

Paper type Research paper

Introduction

In late 2022, the educational landscape was upended by the public release of ChatGPT, a generative AI system trained on vast amounts of textual data. Within a week, over one million users had experimented with the tool (Heaven, 2023), prompting headlines ranging from the alarmist to the optimistic. “The college essay is dead,” declared one former professor (Marche, 2022), while others saw in ChatGPT a new opportunity to “spark new ideas or arguments” (Ceres, 2023). Whether viewed as a disruptor or a digital assistant, ChatGPT has rapidly become embedded in student workflows, blurring the lines between study aid, tutor and ghostwriter.

ChatGPT is a form of a generative pretrained transformer (GPT), capable of producing human-like text, answering questions and supporting a variety of academic and creative tasks (Floridi and Chiriatti, 2020). Its conversational interface and ability to generate full-length essays, summaries or citations have made it particularly appealing in educational contexts, raising urgent questions around academic integrity, critical thinking and the evolving definition of “original” student work (Baidoo-Anu and Owusu Ansah, 2023; Dehouche, 2021; Zhai, 2022). Concerns have emerged around plagiarism – both intentional and unintentional – as well as the possibility that overreliance on AI tools may erode students’ cognitive independence (Miwa *et al.*, 2015).

While much of the initial discourse surrounding ChatGPT centered on concerns about plagiarism, misinformation and the future of assessment (Dehouche, 2021; Qadir, 2023), a growing number of educators have adopted more pragmatic or optimistic stances. Some have compared AI chatbots to calculators for writing – tools that, when used transparently and ethically, can enhance rather than replace learning (Ceres, 2023). Others have urged schools to neither ban nor blindly embrace ChatGPT, but instead to develop instructional models that integrate AI use into reflective, skill-building tasks (Baidoo-Anu and Owusu Ansah, 2023).

In this shifting landscape, understanding how students themselves are using these tools – and for what purposes – is vital. The majority of research to date has focused on hypothetical use cases, institutional policies or theoretical implications. Less is known about the everyday, self-directed practices of students who encounter generative AI in the context of real academic demands. This study seeks to fill this empirical gap by exploring ChatGPT adoption among first-year college students, relying upon their descriptions of how they perceive and use this generative AI tool. Drawing on 38 voluntary reflections from a Fundamentals of Technology course, this paper explores how students understand and engage with AI in support of their learning. Using inductive thematic analysis (Braun and Clarke, 2006), this paper highlights five major themes in student use: the purposes for which AI is used by students, perceived benefits of use, challenges, overall attitudes and the specific tools used. Rather than examining AI use through the lens of policy compliance or technical capacity alone, this study prioritizes students’ authentic voices, offering a grounded perspective on the evolving relationship between AI and undergraduate learning.

While much attention has been paid to institutional responses and technological capabilities of AI, less focus has been placed on how student voices reveal real-world ethical dilemmas in everyday learning. The ways students use AI tools – to write, solve problems or generate ideas – are not ethically neutral. These actions raise pressing questions about fairness, academic honesty and access: Does using AI create an unfair advantage? Are all students equally equipped to benefit from these tools? How does reliance on AI reshape student agency and responsibility? By foregrounding students' lived experiences, this study offers a grounded entry point into these broader ethical debates.

Literature review

Pitfalls and benefits of ChatGPT's use in student writing

Educators and researchers have expressed both enthusiasm and apprehension about the growing presence of AI tools like ChatGPT in academic writing (ElSayary, 2023; Arguello *et al.*, 2024). On one hand, these tools are seen as offering new forms of support for students; on the other, they raise important concerns related to academic integrity, accuracy and skill development. Understanding these perceptions is key to evaluating the implications of ChatGPT's integration into educational contexts.

A major concern regarding the use of ChatGPT as an academic writing tool is that it is capable of generating full essays, potentially introducing a new form of plagiarism into the classroom (Dehouche, 2021). This anxiety is compounded by the fact that, despite ChatGPT's power to draw from a massive data set sourced from books, articles, scientific journals, Wikipedia and other sources, it can provide users with inaccurate or misleading information (Weidinger *et al.*, 2021).

Building on this concern, some instructors have raised hypothetical scenarios in which students might engage in plagiarism through the use of ChatGPT. One risk is that students may submit entire essays or assignments composed entirely of AI-generated text while presenting the ideas as their own (Pophal, 2023). Another concern is that, even when used as a writing aid, ChatGPT may produce responses containing phrases that closely resemble existing published content. "Similar or exact phrases already exist in the larger body of written language, and writers need to be mindful of how this impacts their work" (Pophal, 2023). These examples illustrate how students may engage in plagiarism – intentionally or unintentionally – through improper or uninformed use of AI-generated text.

Despite these concerns, some scholars have noted the potential benefits of ChatGPT as a resource for students. The AI-powered application has been described as having "the potential to offer a range of benefits, including increased student engagement, collaboration and accessibility" (Cotton *et al.*, 2023, p. 1). Other research highlights ChatGPT's usefulness in academic writing, including its ability to summarize lengthy research articles, automate the retrieval of publication details (such as source article publication dates) and generate potential research questions based on a given topic (Dergaa *et al.*, 2023). These concerns highlight the possibility that students could unintentionally or deliberately plagiarize when using AI tools, though further research is needed to assess how frequently these scenarios occur in practice.

Global trends in student ChatGPT use

As generative AI platforms such as ChatGPT grow in popularity among students and continue to embed themselves into academic workflows, a growing body of research has begun probing how and to what extent students are engaging with these tools. While much of the discourse stemming from the launch of ChatGPT in 2022 was focused on US and UK contexts, studies from across the globe have provided insight into students' evolving

relationships with AI across cultural and educational settings. These studies not only highlight the practical and ethical dimensions of student use, but also reveal a shared tension between perceived benefits and underlying concerns – paralleling many of the themes found in the present research.

A recent study found that more than half of university students in Croatia that responded to their survey report using ChatGPT to complete written assignments (Croatia, Črček and Patekar, 2023). Most of these users reported using it to generate ideas, summarize research, paraphrase ideas and proofread. Students consistently reported that generating ideas was the most ethically acceptable form of use, while full-text generation was more ethically ambiguous – though this did not necessarily deter them from using ChatGPT to write parts or even entire assignments. Notably, few students disclose their use of the tool to their instructors, and only a small proportion reported that their use has been detected.

Such patterns have been echoed in a 2023 study by Ngo in which 200 university students in Vietnamese were surveyed and interviewed regarding their ChatGPT use in academia. These student users perceived ChatGPT as a resource that allowed them to use their time more efficiently, probe difficult to understand concepts and clarify information. Praise was given to the tool's ease of use and ability to provide responses tailored to the needs and preferences of the student user. Conversely, students expressed concern and frustration regarding ChatGPT's inability to cite sources accurately, its tendency to fabricate information and citations and its inflexibility with idiomatic language. Several participants noted that they perceived ChatGPT as being helpful for "supporting thinking," but highlighted the necessity of self-regulation, critical thinking and guidance from their academic institutions to ensure the ethical application of generative AI in the academic setting.

A semester-long study by Song *et al.* (2024) at a U.S. university explored ChatGPT's integration into an undergraduate medical imaging course using both general and custom GPT-4-based applications. Their findings demonstrated that approximately 65% of students incorporated ChatGPT into their academic workflows, using it most often for clarifying course content, assistance with programming and support in completing homework assignments. Students noted the tool's interactivity and convenience, yet expressed anxieties surrounding answer accuracy. Their study also revealed that how instructors address ChatGPT use in their classrooms impacts adoption rates of the tool. In a course where the instructor noted the usefulness and sophistication of ChatGPT, student adoption was 87.5%. In contrast, in a course where the instructor cautioned students about the pitfalls of the use of the tool when completing academic assignments, the rate of adoption dropped to 38.2%. These findings suggest that fostering adoption is not solely reliant on generative AI's inherent capabilities, but are also dependent upon how these tools are framed by instructors.

Using latent profile analysis, Stojanov *et al.* (2024) identified five distinct student reliance profiles based on self-reported use of ChatGPT across 13 academic tasks. These ranged from "versatile low reliers" and "knowledge seekers" to "assignment delegators," the latter frequently using the tool to compose large portions, or even the entirety, of written assignments. The study highlights how reliance on ChatGPT is shaped not only by task demands, but also by students' AI literacy, achievement goals and critical thinking dispositions.

A large-scale study by Ibrahim *et al.* (2023) expands this global picture through the combining of performance testing, surveys and cross-cultural analysis. Comparing ChatGPT's responses to actual student work across 32 university courses in disciplines ranging from public policy to computer science, the researchers found that the tool's performance was equal to or better than student submissions in 38% of cases.

Simultaneously, surveys of over 1,500 students and educators from five countries revealed a growing rift: most students reported plans to begin or continue use of ChatGPT, citing skill-building and time savings, while most professors across the same countries indicated they would treat its use as plagiarism. The study also highlighted sharp disparities in access and motivation – students from working-class backgrounds and non-native English speakers were more likely to use ChatGPT and to view it as a tool for leveling the academic playing field. These tensions were further complicated by technological limits: common AI-text detectors such as GPTZero and OpenAI's own classifier frequently misidentified both human and AI-written work, especially after minimal obfuscation, underscoring the practical and ethical challenges institutions now face.

Drawing upon these international findings, [Farhi et al. \(2023\)](#) conducted a comprehensive survey of 388 university students in the United Arab Emirates. The research revealed that ChatGPT significantly shaped students' perspectives on both its practical affordances and the ethical dilemmas it presents. While many respondents emphasized the tool's potential to enhance productivity and written communication skills, they also expressed concerns regarding its possible adverse effects on critical thinking, the development of original thought and overall academic integrity. These findings underscore the complex and often paradoxical relationship students are forming with generative AI, marked by enthusiasm for its capabilities and apprehension about its unintended consequences.

A survey by [Baek et al. \(2024\)](#) involving over 1,000 US college students and found that use of ChatGPT was not only driven by students' academic needs, such as the completion of assignments or studying for exams, but also by social and institutional elements. Older students and non-native English speakers were more likely to use ChatGPT in their academic workflows, particularly for writing tasks, while institutional policy was a major determinant of whether students felt comfortable engaging with generative AI tools in the academic setting. Thematic analysis revealed deep divides in perception: some students saw ChatGPT as empowering and time-saving, while others worried it would erode creativity, enable plagiarism or displace future jobs – especially in STEM and writing-intensive fields. The study's authors emphasize the need for equitable, student-informed approaches to AI integration, arguing that ChatGPT is not a neutral tool but one that reflects and reinforces broader social structures.

Complementing survey and classroom-based studies, [Mogavi et al. \(2024\)](#) conducted a large-scale qualitative content analysis of over 6,000 social media posts to examine early adopters' perceptions of ChatGPT across educational contexts. The study identified a complex emotional landscape: while many users celebrated ChatGPT's ability to boost productivity, self-efficacy and access to on-demand learning, others voiced concerns about overdependence, shallow engagement and the erosion of critical thinking. The findings underscore how public discourse around AI in education mirrors the ambivalence seen in formal student reflections – highlighting both the promise and the pedagogical risks of generative AI.

Taken together, these studies begin to illuminate a global pattern in how students are navigating the challenges and benefits of algorithm-assisted learning. In the absence of clear instructional policies, students appear to be carving out informal norms around when and how to use generative AI. This underscores the need for proactive guidance from educators and institutions – guidance that is attuned not only to the technical capabilities of AI, but also to the realities of how students engage with it in the context of real academic tasks. In viewing these patterns through a cross-cultural lens, the present study contributes to the larger conversation about students' motivations, agency, digital literacy and ethical awareness in the age of AI. Building on these international patterns, this study seeks to

deepen understanding of how students ethically and academically engage with AI tools through self-reported reflections in a supportive classroom setting. Table 1 presents a comparative summary of recent international studies that explore how students engage with ChatGPT in academic settings, highlighting shared benefits, concerns and contextual influences.

ChatGPT and student cognitive load

According to Crawford *et al.* (2023), essay writing can be a particularly fraught endeavor for students as “Writing is already a complex activity but paired with deadlines, other academic pressures and stress, students undertake a substantial cognitive load when completing written-based assignments” (p. 7). This cognitive load, as defined through Cognitive Load Theory, refers the amount of working memory an individual is able to dedicate to a given task (de Jong, 2009). As such, “cognitive capacity in working memory is limited, so that if a learning task requires too much capacity, learning will be hampered,” (de Jong, 2009, p. 105). From this perspective, effective learning occurs when a balance is struck -where the learner is cognitively challenged but not overwhelmed by tasks complexity or information overload.

By assisting students with potential research questions, summaries of complex research articles and other academic research procedures, ChatGPT “could free students from the demand for cognitive resources to reach a state of flow and become fully immersed in the task.” It is within this state of flow that a student “might feel that time passes quickly with

Table 1. Global studies on student ChatGPT use

Author(s) and year	Country/ Region	Methodology	Key findings	Themes
Črček and Patekar (2023)	Croatia	Survey	Over 50% used ChatGPT for assignments; idea generation seen as most ethical; low disclosure to instructors	Ethics, usage patterns, academic integrity
Ngo (2023)	Vietnam	Survey and interviews	Valued for efficiency and clarification; concerns about citation accuracy and information fabrication	Learning aid, critical thinking, self-regulation
Song <i>et al.</i> (2024)	USA	Classroom study	65% usage; instructor attitudes significantly influenced adoption rates	Instructor influence, adoption patterns
Stojanov <i>et al.</i> (2024)	New Zealand	Latent profile analysis	Identified five user profiles; usage linked to AI literacy and achievement goals	User typologies, AI literacy, motivation
Ibrahim <i>et al.</i> (2023)	Multi-country	Performance testing and surveys	ChatGPT outperformed students in 38% of tasks; disparities in access and perceptions noted	Performance comparison, equity, detection challenges
Farhi <i>et al.</i> (2023)	UAE	Survey	Recognized productivity benefits; concerns about critical thinking and academic integrity	Productivity, ethical concerns
Baek <i>et al.</i> (2024)	USA	Survey and thematic analysis	Usage influenced by age, language proficiency and institutional policies; mixed perceptions	Demographics, policy impact, perception divide
Mogavi <i>et al.</i> (2024)	Global	Social media content analysis	Mixed emotions: productivity gains vs. concerns over overdependence and critical thinking erosion	Public discourse, engagement, AI risks

Source(s): Authors’ own compilation based on cited studies

little effort; the act of writing may be enjoyable,” (Csikszentmihalyi, 2008). Though research and brainstorming are critical elements of academic writing that cannot yet be replaced with tools such as ChatGPT, further research may determine whether such AI tools can decrease cognitive load leading to better overall student writing outcomes.

Despite the potential for ChatGPT to reduce the cognitive load for students engaged in academic writing assignments, it is important to note that there are dangers associated with overreliance on AI writing tools as a means to reduce cognitive load. While advanced technologies such as AI powered writing applications can assist writers in brainstorming and summarizing large bodies of research (Dergaa *et al.*, 2023), overreliance upon applications such as ChatGPT may lead to the atrophying of student skills. As echoed by Miwa, Kojima, and Terai (2015), “We often recognize negative secondary effects of the overuse of such systems (e.g. difficulty in memorizing maps due to daily usage of car navigation systems, or difficulty remembering the accurate spelling of words because of using a word processor with spell checker software)” (p. 66). While scholars have raised concerns that frequent reliance on AI writing tools may weaken students’ independent thinking and research skills, limited research has examined how students actually use these tools. This study aims to investigate students’ usage patterns to inform strategies that support the effective and ethical integration of AI in academic settings.

Methodology

This study used a qualitative case study approach to examine how first-year college students interact with artificial intelligence (AI) tools in educational contexts. The investigation was motivated by growing concerns that AI use may negatively affect student learning outcomes and academic integrity. In response, some institutions – such as public schools in New York – have moved to restrict the use of generative AI on campus (Ceres, 2023). However, given the lack of reliable detection tools and the nuanced ways students engage with AI, enforcement remains a challenge.

The data analyzed in this study were not initially collected for research purposes but rather emerged from a classroom-based reflection activity in a Fundamentals of Technology course during the Fall 2024 semester. The course encourages students to explore emerging technologies, including artificial intelligence. As part of an in-class engagement activity, students in three sections of the course (total enrollment = 82) were invited to voluntarily reflect on their use of AI tools in everyday or academic contexts. Participation was incentivized through a small amount of extra credit, and students were informed that they were free to reflect openly on their experiences. The open policy on AI use within the course – as long as students disclosed when it was involved – further reduced concerns about judgment or academic penalties and supported honest self-reporting. A total of 38 student responses were collected and later repurposed for qualitative analysis to better understand student engagement with AI in a real-world educational setting.

Data analysis

The responses were analyzed using an inductive thematic analysis approach. Although the data were not initially collected for research purposes, the reflections were sufficiently rich and relevant to support structured qualitative analysis. All identifying information was removed to preserve student privacy, and the analysis was conducted with the intent of improving our understanding of how students engage with AI in authentic learning environments. No predefined codebook was used. Instead, the analysis was grounded in the data itself, following Braun and Clarke’s (2006) six-phase process:

- (1) familiarization with the data;

- (2) generating initial codes;
- (3) searching for themes;
- (4) reviewing themes;
- (5) defining and naming themes; and
- (6) producing the final report.

Each student response was treated as a unit of analysis. Through open coding, patterns were identified in how students described their use of AI, their motivations, perceived benefits, challenges and overall attitudes. These codes were then organized into broader themes reflecting common experiences and perspectives. This inductive, bottom-up approach ensured that the analysis remained grounded in students’ authentic voices.

Initial coding and theme development were conducted manually and organized using Microsoft Excel. The software was used to document codes, track theme frequency and organize representative examples from student responses. This helped ensure transparency and consistency during the coding and thematic review process.

Findings

Overview of emergent themes

Thematic analysis of 38 student reflections revealed five major themes related to their engagement with artificial intelligence (AI) tools: (1) Purpose of AI Use, (2) Perceived Benefits of AI, (3) Challenges with AI, (4) Attitudes Toward AI and (5) AI Tool Mentions. These themes emerged through inductive coding, using Braun and Clarke’s (2006) thematic analysis approach, grounded in students’ own words and experiences. Each theme includes a summary table of associated codes, followed by a narrative synthesis of key patterns and illustrative quotes.

As shown in Table 2, the most frequently observed student behaviors centered on Problem-Solving Support (25 mentions) and Writing Support (20), which aligned with the broader themes of Cognitive Strategy and Academic Productivity, respectively. Other common uses included Idea Generation (16) and Math/Technical Help (10). Less frequent but still meaningful patterns emerged under themes like Exploratory Use (e.g., creative applications or testing prompts) and Critical Use or Trust Issues (e.g., privacy or accuracy concerns). These frequency counts helped prioritize the structure of the thematic discussion below.

Table 2. Summary of code and theme frequencies

Code	Frequency	Theme
Problem-Solving support	25	Cognitive strategy
Writing support	20	Academic productivity
Idea generation	16	Creative and inspiration
Math/chemistry/technical help	10	STEM support
Research assistance	10	Information seeking
Study aid	8	Cognitive strategy
Creative use	5	Creative and inspiration
Efficiency/Speed	5	Academic productivity
Caution/Concern	3	Critical use or trust issues
Curious experimenter	2	Exploratory use

Source(s): Authors’ own work (inductive coding of student reflections)

Theme 1: Purpose of artificial intelligence use. Students most frequently reported use of AI for Problem-Solving Support (25 mentions), writing assistance (20) and idea Generation (16). Problem-solving included clarifying math problems, Excel steps or complex tasks. Writing support involved grammar editing, APA formatting and structural refinement. Students also described using AI for study preparation, research help and creative projects, revealing a diverse set of academic and purposes.

These findings suggest that students are using AI tools similarly to how they would use writing centers, peer support or academic help resources – for immediate, task-specific assistance. See [Table 3](#) for a summary of key use cases, descriptions and student examples.

Theme 2: Perceived benefits of artificial intelligence. This theme captures how students described the value of using AI, often linked to improve learning, productivity, comprehension and writing confidence. [Table 4](#) summarizes the key benefit categories, along with subcodes, descriptions and representative students quotes.

Five students specifically cited Efficiency/Speed, noting that AI helped them complete assignments faster or structure tasks more effectively. Others mentioned that AI improved their understanding of difficult concepts, especially in subjects like math and chemistry. For example, one student said, “It explained steps better than my teacher.” AI also served as a confidence booster and writing enhancement tool, helping students feel more secure in their works. Although fewer students reflected deeply on benefits, the responses suggest that AI plays a supportive role in easing workload and clarifying expectations, serving traditional tutoring functions. These benefits aligned with findings in Theme 1, suggesting that students view AI as a convenient, on-demand academic resource.

Theme 3: Challenges with artificial intelligence. Despite overall positive sentiment, a few students raised concerns about limitations and risks of using AI. These included issues of trust, overreliance, data privacy and AI-generated content misaligning with their voice or intent. Specific codes, definitions and examples of these challenges are summarized in [Table 5](#).

Table 3. Purpose of AI use: codes, subcodes and examples

Code	Subcodes	Description	Examples
Writing support	Grammar, formatting, APA, structure	Used for proofreading, formatting and learning citation or organization styles	“Used ChatGPT to help me format my paper”
Math/technical help	Excel, equations, formulas	Used to solve math problems or explain technical steps in platforms like excel	“Helped me figure out formatting techniques for excel”
Study aid	Study guides, exam prep	Used to define terms, generate summaries or organize topics for review	“Generated a study guide for my US History exam”
Creative use	Stories, scripts, voice generation	Used for artistic, humorous or experimental projects	“Used AI to help me write a musical plot”
Idea generation	Brainstorming, titles, research topics	Used to generate ideas when stuck or unsure where to start	“Helped me brainstorm a research paper topic”
Research assistance	Sources, citations, topic scoping	Used tools like perplexity or ChatGPT to gather initial research and references	“Perplexity showed me sources for my topic”
Problem-Solving support	Task support, alternative steps	Used to break down problems and learn new approaches	“It explained steps better than my teacher”

Source(s): Authors’ own work (inductive coding of student reflections)

Table 4. Perceived benefits of AI: codes, subcodes and examples

Code	Subcodes	Description	Examples
Efficiency/ Speed	Time-saving, quick results	AI accelerated tasks like editing, brainstorming or checking work	“It gives faster results than I could on my own”
Improved understanding	Explanations, clarification	AI helped students understand concepts more clearly than class materials	“Helped me understand math better”
Confidence booster	Validation, reassurance	AI reassured students their writing or answers were on the right track	“It gave me confidence in my essay structure”
Writing improvement	Clarity, grammar, polish	AI enhanced the quality and correctness of written work	“Helped me with grammar and better word choice”
Perspective shifting	New approach, alternative views	AI helped students consider different perspectives or solutions	“Made me think of problems in a new way”

Source(s): Authors’ own work (inductive coding of student reflections)

Table 5. Challenges with AI: codes, subcodes and examples

Code	Subcodes	Description	Examples
Misunderstood prompts	Vague or off- topic responses	AI misunderstood unclear or complex prompts	“Had to rephrase my question several times”
Overwriting or over-fixing	Unwanted tone/ structure changes	AI altered content too much or in a way not aligned with student voice	“It made my sentence sound too professional”
Information trust issues	Bias, inaccuracy, hallucination	Students questioned the objectivity or truth of AI-generated content	“It seemed like it was pushing an agenda”
Over-Reliance concern	Cheating worries, loss of learning	Concern that AI shortcuts reduce genuine learning or effort	“I used to depend on it too much and realized I didn’t learn anything”
Privacy concerns	Fear of misuse or exposure of data	Students hesitant to input personal or academic info	“I worried it would leak my information”

Source(s): Authors’ own work (inductive coding of student reflections)

For example, one student stated, “It made my sentence sound too professional,” pointing to the issue of overwriting or overfixing. Others noted privacy concerns, expressing discomfort inputting academic or personal data into AI systems. One student wrote, “I worried it would leak my information.” Others flagged issues with inaccuracy or bias, sharing that AI responses sometimes felt incomplete, vague or even ideologically slanted. Although these reflections came from a small portion of the sample (three students), they indicate that some users are beginning to critically assess the ethical and pedagogical implications of AI – especially in relation to academic voice, learning authenticity and data privacy.

Theme 4: Attitudes toward artificial intelligence. This theme reflects the broader mindset students expressed toward AI use- whether cautious, enthusiastic, skeptical or reluctant. Code labels, descriptions and representative quotes for these attitudes appear in [Table 6](#).

Attitudes ranged from Curious Experimenters (2 mentions) to Skeptical but Open users. Some adopted AI only when necessary, expressing a preference for figuring things out independently. Others approached AI as an active learning aid, intentionally using it to scaffold understanding. These responses suggest most students are pragmatic and intentional, not blindly dependent.

Table 6. Attitude toward AI: codes, subcodes and examples

Code	Subcodes	Description	Examples
Curious experimenter	Entertainment, testing prompts	Casual use for fun or exploration without academic goals	"I mess around with ChatGPT just to see what it says"
Skeptical but open	Hesitant use, limited trust	Use AI occasionally or with caution	"I only use it when stuck; I prefer solving things myself"
Active learner	Self-driven, learning-focused	Uses AI as a scaffold to improve learning, not just for answers	"I used it for ideas and then rewrote everything myself"
Resistant or minimal user	Avoidance, rare usage	Doesn't use AI unless required or directed by a class	"I don't usually use AI, only for this class assignment"

Source(s): Authors' own work (inductive coding of student reflections)

This theme reflects the broader mindsets students expressed toward AI use – whether cautious, enthusiastic, skeptical or reluctant. Some identified as Curious Experimenters, using AI casually for entertainment or to test prompts ("I mess around with ChatGPT just to see what it says"). Others were skeptical but open, saying they only used AI when stuck and preferred solving problems independently. Students with an active learner mindset used AI as a learning aid, not just an answer generator ("I used it for ideas and then rewrote everything myself"). A few were Resistant or Minimal Users, only using AI when assigned ("I don't usually use AI, only for this class assignment"). These responses suggest that most students are pragmatic and intentional, rather than overly dependent, adapting their AI use to fit personal learning preferences.

Theme 5: Artificial intelligence tools used. This theme catalogs the platform students reported using, along with academic or creative functions they served. [Table 7](#) summarizes the range of AI tools mentioned by students, with their specific use cases and functions.

ChatGPT was the most widely used tool, appearing in nearly every use case – from writing and grammar to brainstorming and solving math problems. Students praised its versatility and accessibility. Grammarly and QuillBot were primarily used for editing and

Table 7. AI tool used

Tool	Subcodes or notes	Description/usage context
ChatGPT	Most frequent tool used	Used for writing, explaining problems, summarizing, generating ideas
Perplexity	Research tool with citations	Preferred for source-backed information gathering
Grammarly	Grammar/spelling aid	Used for final-stage proofreading
QuillBot	Paraphrasing and sentence polishing	Used to reword content while retaining original meaning
Photomath	Math-specific problem solver	Used for solving and understanding math equations
Gemini	AI assistant (Google's tool)	Used for general assistance and question answering
Chat.AI app	Mobile app with camera input	Used for capturing questions via image input
Voice AI	Voice generation	Used to clone or synthesize voice for creative or humorous purposes
Microsoft copilot	Integrated assistant	Valued for source citation and structured support in research

Source(s): Authors' own compilation from student reflections

sentence polishing, especially at the final stage of writing. For subject-specific needs, Photomath helped solve equations, and Perplexity was preferred for research due to its citation feature. A few students mentioned Voice AI, Chat.AI and Microsoft Copilot, indicating emerging exploration beyond mainstream platforms. This diversity highlights students' ability to match tool capabilities with task demands.

Summary of findings

The analysis of 38 student responses reveals that AI was most frequently used for writing assistance, problem-solving and idea generation. The themes of Cognitive Strategy and Academic Productivity were dominant, while fewer students reflected critically on privacy or overuse concerns. These patterns suggest a growing normalization of AI in educational settings – with many students viewing it as a flexible extension of traditional support systems such as tutoring or writing centers. However, the relatively low frequency of cautionary reflections suggests that educators should continue to foster dialogue about ethical use, privacy and critical engagement as AI tools become more normalized in educational contexts.

Discussion/limitation/future study

The analysis of 38 student responses reveals that students primarily use AI as a tool to support academic tasks – most commonly for writing assistance and problem-solving in subjects such as math and chemistry. The second most frequent use involved idea generation and information seeking, where AI was often perceived as more productive than traditional search engines. Students relied on AI for brainstorming paper topics, formatting essays and gaining explanations for difficult concepts. These findings reflect dominant themes of Cognitive Strategy and Academic Productivity, highlighting the growing role of AI as an on-demand learning assistant.

ChatGPT was the most frequently mentioned tool, cited in nearly every use context – from math and writing support to idea generation. Other tools such as Grammarly and QuillBot were used for polishing written work, while Perplexity was valued for its citation-backed research support. Mentions of tools like Photomath, Voice AI and Microsoft Copilot illustrate a growing diversification in students' tool preferences and adaptability to task-specific needs.

Students generally reported positive experiences with AI. However, a smaller number expressed concerns related to overreliance, privacy or dissatisfaction with overly polished AI-generated responses. For example, some students noted that AI misunderstood vague prompts or rewrote their sentences too formally, which made the writing feel less authentic. These reflections highlight digital literacy challenges, particularly around prompt engineering and maintaining one's academic voice. A few students also expressed hesitation in entering personal information, citing fear of data misuse. While these concerns were less common, they align with broader conversations in higher education about the ethical risks and limitations of generative AI.

The findings of this study echo and extend several patterns highlighted in prior global research summarized in [Table 1](#). For example, similar to [Črček and Patekar \(2023\)](#) and [Farhi et al. \(2023\)](#), students in this study expressed ethical concerns about ChatGPT, particularly regarding academic integrity and disclosure. The use of ChatGPT as a learning aid, noted by [Ngo \(2023\)](#), was also evident in participants' reported reliance on the tool for clarifying assignments and generating ideas. However, diverging from the typologies identified by [Stojanov et al. \(2024\)](#), this study found a more fluid spectrum of user behavior rather than distinct profiles, possibly due to the broader disciplinary scope of participants. Furthermore,

this study adds nuance to the influence of institutional policies reported by Baek *et al.* (2024), highlighting that students' perceptions of policy clarity were often shaped by informal peer norms rather than formal guidelines. In this way, the current findings both confirm global patterns and offer new insights into student rationales and socioacademic contexts of ChatGPT use.

Implications for faculty and instructional design

The widespread reliance on AI – particularly for tasks like drafting, editing and organizing written work – raises important questions about the structure and purpose of traditional assignments. As students increasingly turn to AI for grammar, formatting and idea development, faculty may need to rethink how to design assignments that maintain academic rigor while integrating AI as a learning tool. For example, rather than prohibiting AI use, instructors could encourage students to reflect on how they use these tools or incorporate AI-assisted drafts into revision-based writing tasks.

Moreover, student frustrations with AI responses suggest a gap in digital literacy – particularly in how to effectively prompt and critically evaluate AI output. This opens an opportunity for educators to introduce basic AI literacy and responsible use into foundational courses. Rather than focusing solely on detection tools or restrictive policies, faculty might prioritize helping students use AI to deepen learning, not bypass it.

Ethical implications of student AI use

Although students largely viewed AI tools as helpful for writing support and problem-solving, the ethical implications of such use warrant deeper reflection. For instance, students with stronger digital literacy or more comfort with AI prompt design may gain efficiency and writing advantages – raising concerns about fairness and academic equity. If AI-enhanced productivity becomes the norm, students without access to these tools or the skills to use them effectively, may be left behind. This dynamic, risks exacerbating educational inequality rather than closing it. Academic integrity also emerges as a blurred boundary. While most students described AI as a scaffold rather than a substitute, the line between legitimate assistance and outsourcing cognitive work remains ambiguous. If students use ChatGPT to generate outlines, suggest citations or rewrite paragraphs, at what point does authorship become shared – or compromised? These questions are especially urgent as institutions refine their definitions of plagiarism in an AI-driven era. Rather than simply promoting or banning AI, educators must consider how instructional design, disclosure norms and access policies shape ethical student behavior.

While this study surfaced several key ethical concerns surrounding AI use in higher education, the list presented here should be understood as preliminary rather than exhaustive. Other dilemmas – such as how students weigh AI explanations against instructor authority, or how instant output may affect time spent on deep learning – were observed but fall outside the scope of this article. Future work may explore these nuances further.

Limitations and future research

This study is limited by its small sample size and course-specific context. Because participation was voluntary and offered for extra credit in a class that encouraged AI use, the sample likely overrepresents students who were positively inclined toward generative AI. Those with strong ethical concerns or discomfort using such tools may have opted out. Furthermore, the open environment may have influenced students to describe their experiences in ways they perceived as acceptable or expected. These factors limit generalizability and suggest the need for broader sampling across diverse instructional

environments and AI policy settings. In addition, while detailed demographic data such as major or language background were not collected, all participants were first-year students enrolled in foundational courses, which likely minimized disciplinary bias. Nonetheless, future research may benefit from examining how other demographic variables, such as gender or linguistic background, shape students' perceptions and uses of AI.

Future research could explore how different instructional designs influence AI use, or how faculty perceptions shape classroom policies around AI. Studies might also examine how students develop AI literacy over time, and which interventions – such as workshops or AI-integrated assignments – effectively promote critical and ethical engagement with AI tools.

Conclusion

In summary, students largely perceive AI tools as accessible resources that support various academic needs, from clarifying concepts to structuring assignments and refining drafts. While concerns about privacy and overreliance were raised by a minority of students, these concerns resonate with ongoing debates in higher education. As AI becomes more embedded in learning environments, the focus should shift from restriction toward thoughtful integration. Ultimately, as with the transition from physical card catalogs to digital databases, educators and researchers must guide students in using emerging technologies wisely, critically and creatively.

While AI tools like ChatGPT are often discussed in terms of productivity and assistance, their use raises complex ethical questions. For instance, when students rely on AI to generate writing, what does this mean for academic honesty? If some students have more AI literacy or access than others, are we creating new forms of inequality? This study seeks to illuminate such questions by analyzing students' own descriptions of how they use – and sometimes question – these technologies in real academic contexts.

References

- Arguello, J., Vázquez-Cano, E. and Vázquez-Toledo, S. (2024), "Analysis of teachers' perceptions of the educational use of ChatGPT", *Revista Complutense de Educación*, Vol. 35 No. 1, pp. 23-36, doi: [10.5209/rced.87580](https://doi.org/10.5209/rced.87580).
- Baek, Y., Kim, J., Chang, S., Li, Y. and Warschauer, M. (2024), "ChatGPT seems too good to be true": college students' attitudes and experiences with generative AI", *Computers and Education*, Vol. 207, p. 105799, doi: [10.1016/j.compedu.2024.105799](https://doi.org/10.1016/j.compedu.2024.105799).
- Baidoo-Anu, D. and Owusu Ansah, L. (2023), "Education in the era of generative artificial intelligence (AI): understanding the potential benefits of ChatGPT in promoting teaching and learning", *Social Education Research*, Vol. 4 No. 1, pp. 108-115, doi: [10.37256/ser.4120232071](https://doi.org/10.37256/ser.4120232071).
- Braun, V. and Clarke, V. (2006), "Using thematic analysis in psychology", *Qualitative Research in Psychology*, Vol. 3 No. 2, pp. 77-101, doi: [10.1191/1478088706qp0630a](https://doi.org/10.1191/1478088706qp0630a).
- Ceres, P. (2023), "ChatGPT is coming for classrooms", Don't panic. Wired, available at: www.wired.com/story/chatgpt-is-coming-for-classrooms-dont-panic/
- Cotton, D. R., Cotton, P. A. and Shipway, J. R. (2023), "Chatting and cheating: ensuring academic integrity in the era of ChatGPT", *Innovations in Education and Teaching International*, Vol. 61 No. 2, pp. 228-239, doi: [10.1080/14703297.2023.2190148](https://doi.org/10.1080/14703297.2023.2190148).
- Crawford, J., Cowling, M. and Allen, K. (2023), "Leadership is needed for ethical ChatGPT: Character, assessment, and learning using artificial intelligence (AI)", *Journal of University Teaching and Learning Practice*, Vol. 20 No. 3, doi: [10.53761/1.20.3.02](https://doi.org/10.53761/1.20.3.02).

- Črček, N. and Patekar, J. (2023), "Writing with AI: University students' use of ChatGPT", *Journal of Language and Education*, Vol. 9 No. 4, pp. 128-138, doi: [10.17323/jle.2023.17379](https://doi.org/10.17323/jle.2023.17379).
- Csikszentmihalyi, M. (2008), *Flow*, HarperCollins.
- de Jong, T. (2009), "Cognitive load theory, educational research, and instructional design: Some food for thought", *Instructional Science*, Vol. 38 No. 2, pp. 105-134, doi: [10.1007/s11251-009-9110-0](https://doi.org/10.1007/s11251-009-9110-0).
- Dehouche, N. (2021), "Plagiarism in the age of massive generative pre-trained transformers", *Ethics in Science and Environmental Politics*, Vol. 21, pp. 17-23.
- Dergaa, I., Chamari, K., Zmijewski, P., Ben, and Saad, H. (2023), "From human writing to artificial intelligence generated text: Examining the prospects and potential threats of ChatGPT in academic writing", *Biology of Sport*, Vol. 40 No. 2, pp. 615-622, doi: [10.5114/biolsport.2023.125623](https://doi.org/10.5114/biolsport.2023.125623).
- ElSayary, A. (2023), "An investigation of teachers' perceptions of using ChatGPT as a supporting tool in education", *Journal of Computer Assisted Learning*, Vol. 39 No. 6, pp. 1622-1632, doi: [10.1111/jcal.12864](https://doi.org/10.1111/jcal.12864).
- Farhi, M., Al-Olimat, H.S., Alghamdi, N.S. and Bera, A. (2023), "An era of ChatGPT as a significant futuristic educational technology: Students' perspectives and concerns", *BenchCouncil Transactions on Benchmarks, Standards and Evaluations*, Vol. 3 No. 2, p. 100091, doi: [10.1016/j.tbench.2023.100091](https://doi.org/10.1016/j.tbench.2023.100091).
- Floridi, L. and Chiriatti, M. (2020), "GPT-3: Its nature, scope, limits, and consequences", *Minds and Machines*, Vol. 30 No. 4, pp. 681-694, doi: [10.1007/s11023-020-09548-1](https://doi.org/10.1007/s11023-020-09548-1).
- Heaven, W.D. (2023), "OpenAI's ChatGPT is everywhere. Here's what we know about it", MIT Technology Review, available at: www.technologyreview.com/2023/02/08/1068068/chatgpt-is-everywhere-heres-where-it-came-from/
- Ibrahim, H., Asim, R., Zaffar, F., Rahwan, T. and Zaki, Y.R. (2023), "Rethinking homework in the age of artificial intelligence", *Scientific Reports*, Vol. 13, p. 12187, doi: [10.1038/s41598-023-38964-3](https://doi.org/10.1038/s41598-023-38964-3).
- Marche, S. (2022), "The college essay is dead", The Atlantic, available at: www.theatlantic.com/technology/archive/2022/12/chatgpt-college-essay-artificial-intelligence/672371/
- Miwa, K., Kojima, K., and Terai, H. (2015), "An experimental investigation on learning activities inhibition hypothesis in cognitive disuse atrophy", *COGNITIVE 2015: The Seventh International Conference on Advanced Cognitive Technologies and Applications*, pp. 66-71. http://personales.upv.es/thinkmind/dl/conferences/cognitive_2015/cognitive_2015_4_30_40103.pdf
- Mogavi, A., Koutsoukos, S., Tang, T. and Qu, Y. (2024), "ChatGPT in education: a blessing or a curse? A qualitative content analysis of social media discussions", *Computers in Human Behavior*, Vol. 148, p. 107935, doi: [10.1016/j.chb.2023.107935](https://doi.org/10.1016/j.chb.2023.107935).
- Ngo, T.T.A. (2023), "The perception by university students of the use of ChatGPT in education", *International Journal of Emerging Technologies in Learning (iJET)*, Vol. 18 No. 17, pp. 4-19, doi: [10.3991/ijet.v18i17.39019](https://doi.org/10.3991/ijet.v18i17.39019).
- Pophal, P. (2023), "ChatGPT: Opportunities and risks related to AI-Generated content", *InfoToday*, March 2, 2023.
- Qadir, J. (2023), "Artificial intelligence and the future of education: Issues and opportunities", *IEEE Access*, Vol. 11, pp. 1542-1555, doi: [10.1109/ACCESS.2022.3224424](https://doi.org/10.1109/ACCESS.2022.3224424).
- Song, X., Zhang, J., Yan, P., Hahn, J., Kruger, U., Mohamed, H. and Wang, G. (2024), "Integrating AI in college education: Positive yet mixed experiences with ChatGPT", *arXiv Preprint arXiv:2407.05810*, doi: [10.48550/arXiv.2407.05810](https://doi.org/10.48550/arXiv.2407.05810).
- Stojanov, A., Liu, Q. and Koh, J.H.L. (2024), "University students' self-reported reliance on ChatGPT for learning: a latent profile analysis", *Computers and Education: Artificial Intelligence*, Vol. 6, p. 100243, doi: [10.1016/j.caeai.2024.100243](https://doi.org/10.1016/j.caeai.2024.100243).
- Weidinger, L., Mellor, J., Rauh, M., Griffin, C., Uesato, J., Huang, P.-S., Cheng, M., Glaese, M., Balle, B., Kasirzadeh, A., Kenton, Z., Brown, S., Hawkins, W., Stepleton, T., Biles, C., Birhane, A.,

Haas, J., Rimell, L., Hendricks, L.A.,[...] Gabriel, I. (2021), "Ethical and social risks of harm from language models", arXiv, available at: <https://arxiv.org/abs/2112.04359>

Zhai, X. (2022), "ChatGPT for next generation science learning", *SSRN Electronic Journal*, doi: [10.2139/ssrn.4331313](https://doi.org/10.2139/ssrn.4331313).

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